

Applying Information and Communication Technology (ICT)

This unit focuses on the practical application of ICT tools to solve problems and meet specific user requirements.

Core Competencies (Assessment Objectives)

- **AO1 (Knowledge & Understanding):** Grasping the fundamental concepts of ICT.
- **AO2 (Application):** Utilizing software to edit, format, and organize information into functional products.
- **AO3 (Analysis & Evaluation):** Critically reviewing outcomes against original project requirements to identify strengths and weaknesses.

2. Theoretical Framework: Data and Context

- **Definitions:**
 - **Data:** Raw, unprocessed elements (numbers, text, sound, images, or animation). In this raw state, it is meaningless.
 - **Information:** Data that has been processed to include context and meaning.
- **Contextual Necessity:** Data often requires specific context to become information.
 - *Example:* An animation of a sphere circling another could be an Earth-Sun model, a company logo, or something else entirely; it is only "information" once the context is established.
 - *Example:* A simple "beep" might be an alarm, a game sound, or an app notification, depending on the context.

3. Data Types and Meaningful Processing

Data Type	Characteristics	Methods to Produce Meaning
Text	Characters (letters/symbols).	Syntax/structure of language.
Numbers	Integers, decimals, currency, percentages.	Calculation, graphs, and charts.
Images	Bitmap or vector graphics.	Surrounding text, metaphor, and semiotics.

Animation	Files like SWF or animated GIFs.	Storytelling, surrounding information, text.
Sound	Audio clips.	Hardware (sound card/speaker) and supplementary data.

4. Software Application Purposes

- **Word Processor:** Used for letters, menus, reports, and essays. Features include pagination, spell check, and paragraph styling.
 - **Database Management:** Used for entering, filtering, sorting, and searching data.
 - **Graphics:** Used for creating, manipulating, and editing images.
 - **Web Authoring:** Used for creating websites and pages.
 - **Presentation:** Used for displaying multimedia content to an audience.
 - **Spreadsheet:** Used for financial budgeting, modelling, and performing calculations.
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5. Design, Layout, and House Style

- **House Style:** A formal standard for layout and writing within an organization, used to maintain a consistent "corporate identity".
 - **Typography:**
 - **Serif:** Includes a small line at the ends of letters (e.g., *Times New Roman*). Best for printed media as the serif helps guide the reader's eye.
 - **Sans Serif:** Without serifs (e.g., *Verdana*). Often better for low-resolution digital screens.
 - **Colour & Contrast:**
 - **Colour Wheel:** Used to find complementary (opposite) colours for high contrast or analogous (adjacent) colours for thematic harmony.
 - **Accessibility:** Designers must account for colour-blindness. For example, never use red and green together, as some users cannot distinguish between them.
 - **White Space:** Balancing empty space is critical. Too much makes content feel isolated; too little makes it appear cluttered and confusing.
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6. Review, Iteration, and Evaluation

Reviewing and Refinement

- **Automated Checking:** Software provides tools like spell and grammar checkers. These may not catch contextual errors (e.g., "there" instead of "their").
- **Proofreading:** Essential as automated tools may suggest incorrect changes.
- **Iterative Development:** Reviewing products against requirements during development to improve quality.

- **Evaluation:** Involves identifying strengths and weaknesses, considering alternative applications, reflecting on skill usage, and assessing if the product met user needs.
- **Evaluation Criteria:**
 - Did the software selection match the task needs?
 - Could the process be more efficient (e.g., through keyboard shortcuts)?
 - Did the product meet the intended audience/purpose requirements?

Chapter Questions

1. Which one of these data types is used to store a photograph?
 - ☐ (A) Image,
 - ☐ (B) Number,
 - ☐ (C) Text,
 - ☐ (D) Sound.
2. Explain one reason why it is important to use white space well when designing a digital product.
3. State one type of data and describe how it can be turned into information.
4. Identify three errors in this text: "How was I supposed to know Mr Singh".
5. Explain why using a house style can improve finished products when multiple documents are created for one organisation.
6. State what is meant by the term 'serif font'.
7. State one drawback of using serif fonts on devices with low-resolution screens.
8. Using the colour wheel, identify and state the name of the colour that contrasts most with red.
9. State one accessibility need relating to the use of colour in product designs.
10. State why it is important to proofread your work to check for errors.